A Survey of Cultural Awareness in LLMs: Text and Beyond

Siddhesh Pawar*, Junyeong Park*, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav,
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Paper **Github**

A Survey of Cultural Awareness in LLMs

- **Review 300+ papers** to provide a comprehensive overview of current research on cultural inclusion in **(multimodal) language models**
- Discuss the: 1) common data sources; 2) alignment strategies; and 3) evaluation methods used for cultural alignment
- Examine the implications for: 1) AI ethics; 2) social science; and 3) the role of HCI (Human-Computer Interaction) research in the development of culturally aware LLMs

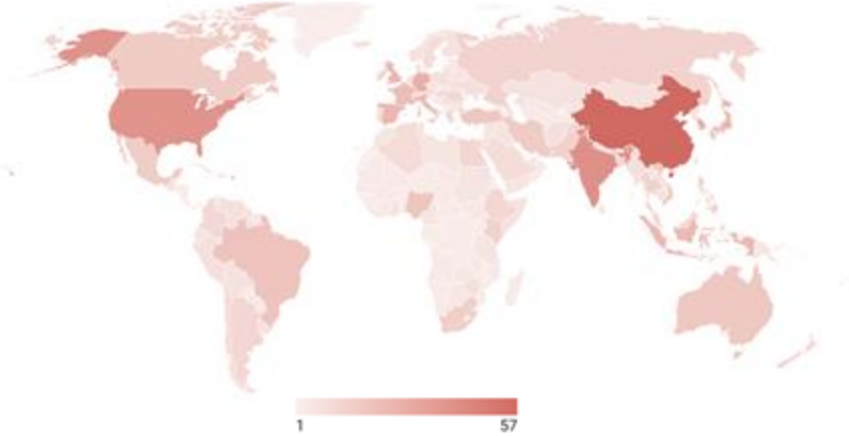
Language and Region Coverage

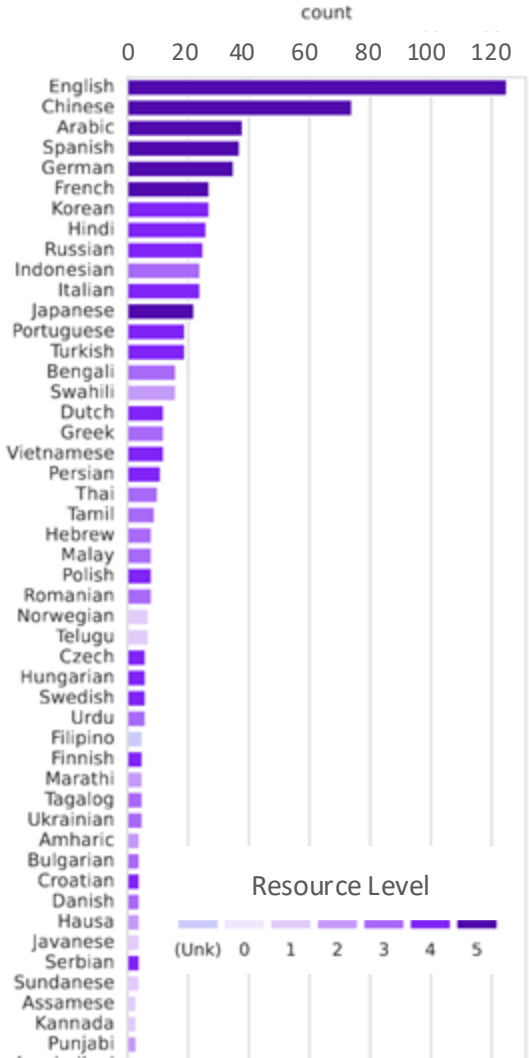
- **Languages and countries** covered by **cultural evaluations across different modalities**

➔ *Expand research on **low-resource cultures and languages***

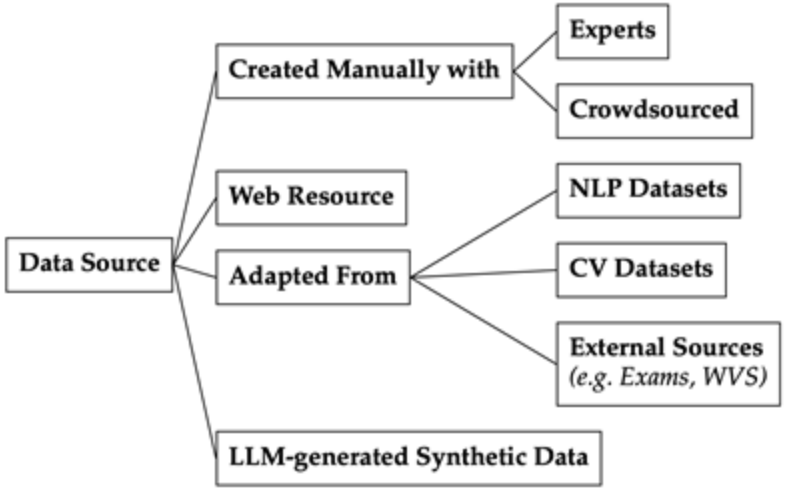
➔ *Approach defining **cultural boundaries** with caution*

- ◆ Countries do not always align with cultural boundaries
- ◆ Some languages span multiple cultural contexts





Dataset Creation for Cultural Inclusion



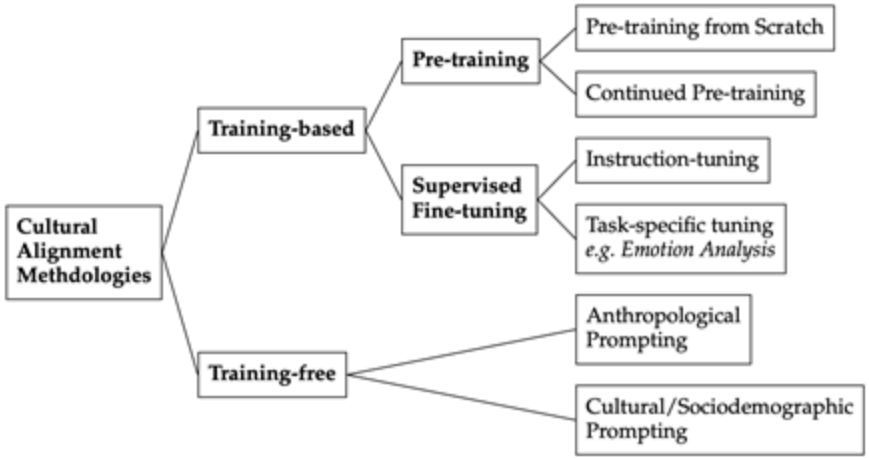
- **Automatic & Model-in-the-loop**
  - Employed for collecting training or instruction data
  - Largely relies on publicly available multilingual large-scale corpora (e.g. Wikipedia, CC100)
- **Semi-automatic: Human-in-the-loop**
  - Manual localization of automatically collected or generated datasets
  - Manually create an initial dataset or guidelines for machines to generate synthetic data
- **Manual creation**
  - Primarily used for generating golden answers in evaluation benchmarks

Cultural Dataset Creation: Takeaways

- ➔ *Ensure **inclusive** cultural representations*
  - ◆ Even within a cultural group, social values may vary based on demographics
  - ◆ Involve diverse and sizable pool of annotators to capture full range of perspectives
- ➔ ***Vary data collection strategies** according to target culture*
  - ◆ Consider technology access of a target culture
  - ◆ e.g. Sudan - low technology penetration; may need to directly recruit natives
- ➔ *Explore **alternative image data collection** method to **mitigate biases in Web images***
  - ◆ e.g. Apprehension bias; people may pose and look differently when they know they are being photographed

Cultural Alignment Methodologies

- Most alignment efforts focus on **aligning LLMs to individual local cultures**, with limited research on developing cross-cultural LLMs
- While training-free methods can improve cultural alignment, the **robustness of sociodemographic prompting remains uncertain** and should be used with caution



{Anthropological framework}

*Imagine you are a **{marital_status} {sex}** from **{region}, {country}.** You are **{age}** years of age and completed **{education}** education level. You consider yourself part of the **{social_class}**. Answer the following question from this perspective. Others will read what you choose; your goal is to convince them it was chosen from the perspective of the persona described above.*

*First, **provide your reasoning based on the upper anthropological framework** then select exactly one option from the options below…*

Anthropological prompting

*"My neighbor is **{nationality}**. My neighbor probably likes listening to …*

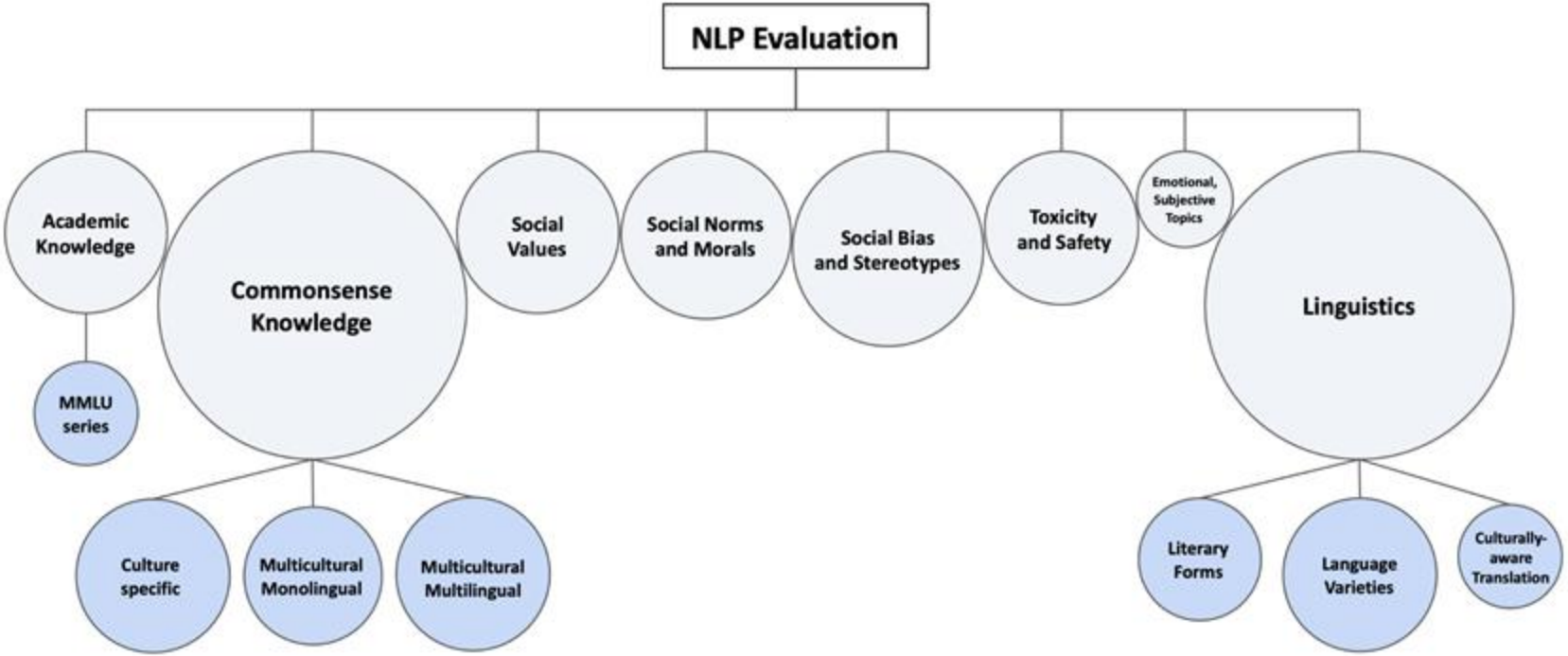
Cultural/Sociodemographic prompting

Cultural Alignment Methodologies: Takeaways

- ➔ ***Balance*** *development of* ***culturally-specific LLMs*** *and* ***comprehensive universal LLMs***
  - ◆ Current research focuses less on creating cross-cultural LLMs
- ➔ *Develop LLMs that can* ***adapt and evolve with cultural change over time***
  - ◆ No culture remains static
  - ◆ While some studies explore historical cultures, there remains a notable gap in adapting LLMs as cultures evolve

Cultural Evaluation Benchmarks: Language

- NLP benchmarks can be categorized into the following **eight domains**

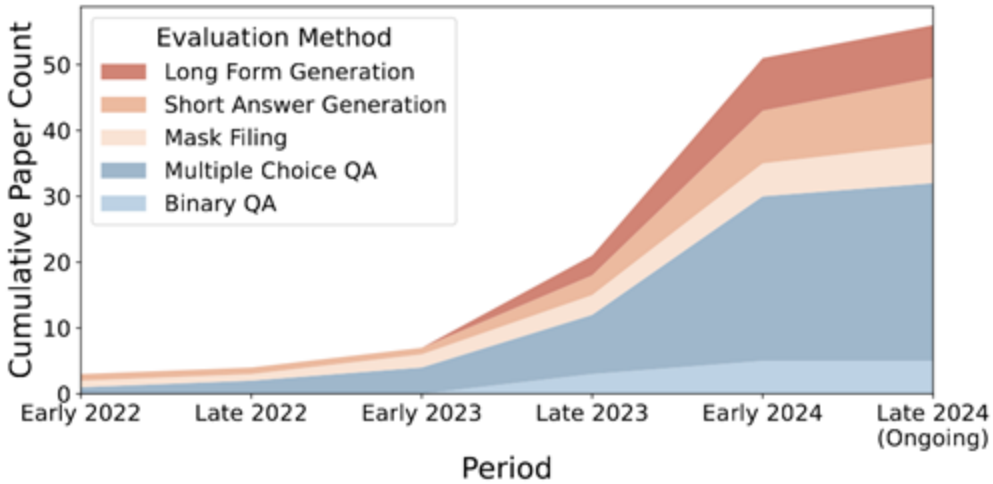


Cultural Evaluation Benchmarks: Takeaways

- Evaluation methods have primarily relied on MCQs, but **generation based approaches are gaining traction**

➔ *Expand cultural evaluation methods to **diverse interaction settings***

- *real-life natural conversation setting*



Culturally-Aware Models across modalities

- **Vision-Language**
  - Visual Question Answering (VQA)
  - Image Captioning
  - Culture-specific Text to Image (T2I)
  - Art-forms related tasks
- **Audio and Speech**
  - Emotion Detection
  - Music Recommendation
- **Video**
  - Emotion Detection
  - Content Moderation

Implications for Social Sciences, and Role of HCI

- Culturally-aware LLMs may **accelerate social science research** by **using it in sociological surveys**, however,
  - Social scientists should be cautious due to potential cultural bias and misrepresentations in LLMs
  - Generative model designers must avoid over-relying on synthetic data, as culturally misaligned LLM outputs can perpetuate a cycle of misrepresentation
- **Role of HCI** in developing culturally-aware LLMs
  - Understand users' expectation from language technologies (across cultures)
  - Assess systemic impact of deploying LLMs in public

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